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Design of an Underactuated Adaptive Robotic Hand with Force Sensing

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***Abstract*— In this paper the design of an underactuated anthropomorphic robotic hand is presented. The hand is highly-underactuated and passively adaptive to the shape of an object, due to a tendon-driven mechanism and torsional spring in each finger joint. Each of five fingers has three DOFs (Degrees Of Freedom), except the thumb which has an additional DOF, for the rotation in its base. The fingers are tendon-driven, actuated with five DC motors, embedded in the palm. Flexion and extension of the little and ring finger are coupled, since they are actuated with a single motor. The middle and index finger are actuated with one motor each, while the thumb has one motor for flexion/extension and another for the rotation in its base. Custom made 3-axis force sensors are integrated at the tip and in the base link of each finger. Force sensors consist of four Hall sensors and a small permanent magnet in the middle, sealed in a silicon base. The design of the robotic hand is highly anthropomorphic and biologically inspired by the human hand.**

***Index Terms*— Robot hand, Hand design, Underactuation, Force sensors, Adaptive hand.**

I. Introduction

Considering the current trend and goal in robotics research community to provide working and living coexistence of robots and humans, it is necessary to provide the robots with a capability to interact with their environment. The hand plays very important role in manipulation, which is the most fundamental task of humanoid robots. Design of an anthropomorphic robotic hand is a very challenging task. The robotic hand is a complex mechatronic device whose design has to provide sufficient dexterity, speed and mass/payload ratio in presence of multiple constraints imposed on its weight, size and packaging. Since our environment and most of the tools and objects are ergonomically designed to fit the shape of human hands it is reasonable to consider bioinspired anthropomorphic structure for the design of a robotic hand which would be capable to manipulate with objects of different shapes and sizes. In order to use robotic hands in unstructured environment they have to possess certain degree of compliance, to deal with possible impacts and collisions, and passive or active adaptability to different object shapes. Also, the hand should be equipped with force sensing to manipulate with objects of different mechanical characteristics, including fragile objects. It is very hard to design a robotic hand able to perform robust grasps of common objects in reliable and repeatable manner, in presence of noise and sensing errors. Despite many research papers and significant progress, this might be still considered an open problem. Human hand has around 20 DOFs and it is characterized with great mobility and dexterity. It is very hard to provide compact and functional design for artificial anthropomorphic hand with so many DOFs, since it would demand great volume for actuators. Therefore, many solutions have been proposed for underactuated robotic hands, where several joints are coupled and actuated with a single actuator.

The rest of the paper is organized as follows. In section II an overview of existing robotic hands is given to present the state of the art in the field. After that, in section III kinematic parameters of the robot hand are given. In the next section mechanical design of the proposed robotic hand is described with an emphasis on the custom made 3-axis force sensor. At the end, conclusion is given with a brief reference to research topics for future work.

II. Overview

After few decades significant progress has been made in the field of artificial hand design, but it is still an open problem with ongoing research. Development of new materials and new types of actuators is also promising for the next generation of robotic hands. Some state of the art solutions for the robotic hand design, characterized by great functionality and dexterity are the Robonaut hand, designed by NASA [1,2], the series of DLR hands of different generations [3-4], The GIFU hand III [5], the UB Hand III [6], Shadow hand [7], Elumotion Hand 2 [8] and many others. Common drawback of these robotic hands is their complex design and high price. Mostly they are still large and heavy, and they are not able to provide large grasp force, compared to the human hand. Robotic hands can be classified based on several criteria.

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Considering the type of actuation, most commonly used type of actuators are servomotors. Vast majority of robotic hands, including the one presented in this paper use this type of actuators [1-6, 8]. Servomotors are suitable for control, but have poor mass/payload ratio. Pneumatic artificial muscles (PAMs) are lightweight and have much better mass/payload ratio than electric motors. PAMs have similar force characteristic as human muscles, but the characteristic is nonlinear. Great advantage of PAMs is their intrinsic compliance, but they need compressed air, which limits the mobility and manipulability of such robotic hands. Several dexterous robotic hands that use PAMs for actuation have been developed [7, 9]. Another solution of the lightweight artificial hand design, with special flexible fluidic actuators, is reported in [10]. An innovative design has been proposed for robotic hand, which uses so called hydraulic cluster actuators [11]. This electro-hydrostatic actuator generates great force, more than 260 N, yet it is backdrivable. Beside above mentioned actuators, some other types of actuators might also be used like: piezoelectric actuators, memory alloys and electro-active polymers.

III. Kinematic Parameters

Robotic hand proposed in this paper has five fingers and its design and kinematic structure are biologically inspired by human hand. Each finger has 3 joints (3 DOFs) which provide the flexion/extension movement of the finger. The thumb has 3 DOFs for flexion/extension and additional DOF for the rotation in its base. Structure of the hand with its kinematic parameters is shown in fig. 1.

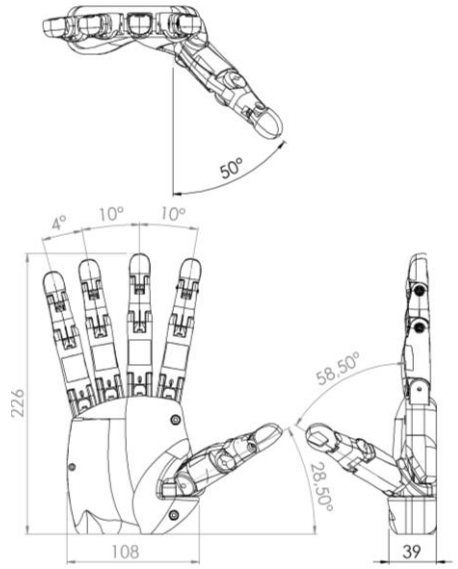


Fig. 1. Structure of the hand and its kinematic parameters

Mechanical design of the finger joints is shown in cross section in fig. 2. There are two radial ball bearings in each joint and a torsional spring. Small lock washers are used to constrain the axial movement of the joint shafts. Flat cables for force sensors and steel tendon for finger actuation are led through finger segments through guiding holes.

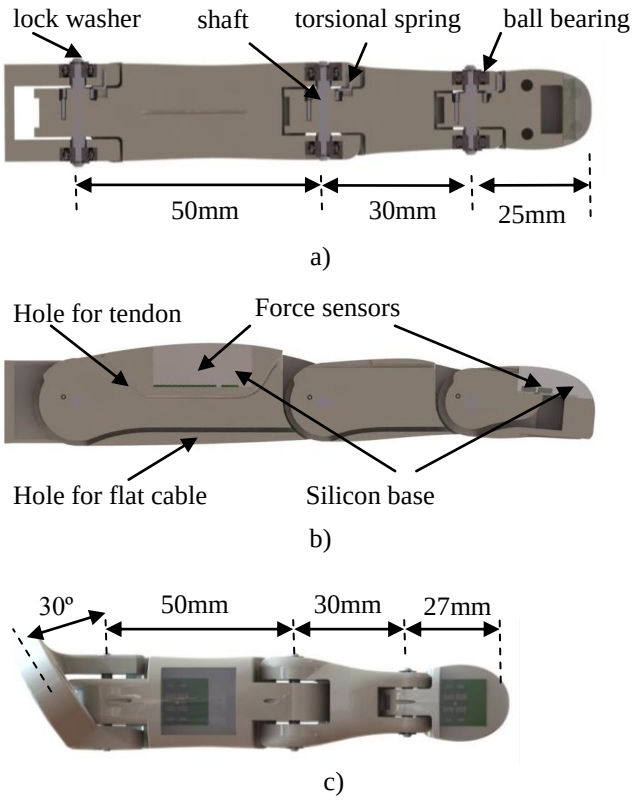


Fig. 2. a) Frontal cross section of the finger with kinematic parameters of the links b) Lateral cross section of the finger c) Kinematic parameters of the thumb

Kinematic parameters of the robot hand and specification of the used actuators are given in table 1.

TABLE I
Design specification of the hand

Dimensions	Total (h/l/w)	226x108x39 mm
	Finger length	105 mm
	Thumb length	107 mm
Weight	Total	980 g
	Finger	31 g
	Thumb	47 g
Operating angles of joints	1st finger joint	90º
	2nd finger joint	90º
	3rd finger joint	90º
	4th joint	55º
Actuator	Weight	22 g
	Size	Ø16 x 25.4 mm
	Torque (nominal)	2.16 mNm
	Speed (nominal)	4900 rpm
Gearhead	Weight	29 g
	Size	26.8 mm
	Torque (nominal)	0.4 Nm

IV. MECHANICAL DESIGN

In order to provide lightweight structure of the hand, all finger segments and the palm are supposed to be realized on a 3D printer. This production technique is suitable for rapid prototyping and allows complex shapes to be realized in order to provide anthropomorphic appearance and compact design. Unlike the most robotic hands which have actuators relocated in the forearm, the hand proposed in this paper has 5 brushed DC motors embedded in the palm. The complete power and control electronics will be also integrated in the palm, which is designed as a shell structure. Therefore, the proposed robotic hand is an independent and compact mechatronic device that can be easily combined with different industrial or anthropomorphic robotic arms. CAD model of the robotic hand is shown in fig. 3.



Fig. 3. CAD model of the robotic hand

Since only 5 motors are used for actuation of 16 joints, the hand is highly underactuated. Flexion/extension movement of the little finger and the ring finger is coupled, since they are actuated with a single motor. Middle finger and the index finger are actuated with one motor, each, while the thumb has one motor for flexion/extension movement and another one for the rotation in its base. Placement of the actuators can be seen in fig. 4. where the interior of the palm is shown.

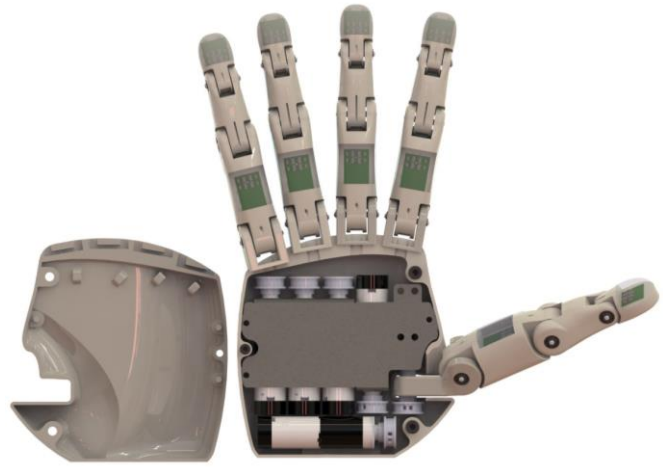


Fig. 4. Actuation system embedded in the palm

Tendon driven mechanism is used for the flexion/extension of each finger and these movements are actuated by the motors dislocated in the palm. Each tendon goes from the pulley on the motor shaft in the palm, through the guiding holes in all finger's segments to the fingertip, where it is fastened. Tendons and springs in joints impose constraints on finger motion and provide coupled motion in finger joints. In order to minimize the friction in the finger joints, small radial ball bearings are integrated in each joint. The rotation in the base of the thumb is also tendon driven. Two pulleys were used for power transmission in the base joint of the thumb. Unlike the other joints, tendon mechanism rotates the thumb joint in both directions, without a spring.

In each finger joint there is a torsional spring. Hence, the finger flexion is realized by DC motor, as a single active element, while the extension is realized by torsional springs, as antagonistic passive elements. The stiffness coefficients of these springs define the trajectory of the finger and the way it bands during unconstrained motion, in the absence of external forces and obstacles. This tendon driven mechanism with springs provides certain passive adaptability to the shape of object that is being gripped. If the first finger segment hits the obstacle, while banding, the second and third segment will continue to bend, until they hit the obstacle or reach their mechanical limits. The concept of passive adaptation to the shape of object is illustrated in fig. 5.

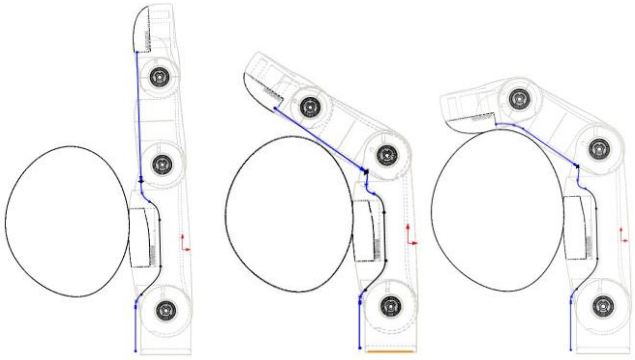


Fig. 5. Passive adaptation of the finger configuration to the shape of object due to the tendon driven mechanism and torsional springs in finger joints.

The robotic hand has the ability of force sensing which can be used to detect whether an object has been grasped or not. Each finger has two custom made 3-axis force sensors integrated in the base link and fingertip of each finger. Since these sensors can measure force intensity in three orthogonal directions, and not just the presence or absence of an object, it is possible to grasp objects of different mechanical characteristics, including fragile objects. Force sensors are based on measuring the change of magnetic field. Force sensor consists of a PCB (Printed Circuit Board) with four Hall sensors mounted in a square formation and a small neodymium magnet in the middle, above the Hall sensors. Both PCB and the magnet are sealed in the viscoelastic base, made of silicon rubber which is integrated in finger segments. When the hand grasps an object, silicon base that is in contact with the object deforms. Due to the viscoelastic deformation of the silicon base permanent magnet changes its position relative to the Hall sensors. This deformation induces the

change in magnetic field, which results in a change of output voltage of the Hall effect sensors. Working principle of this sensor is illustrated in fig. 6.

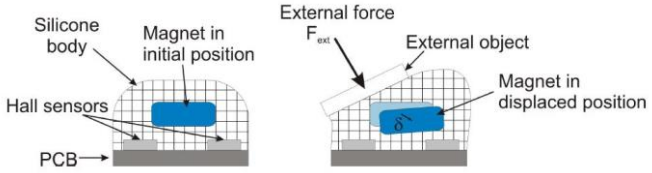


Fig. 6. Working principle of the 3-axis force sensor

Silicon parts have high friction coefficient and intrinsic passive compliance which are desirable for grasping. Precise 6-axis force/torque sensor was used to form the training set for the neural network, which was used for signal processing of the measured data [12]. Flat cable is guided from each force sensor to the electronics in the palm through guiding holes in finger segments.

Many empirical studies of human grasping have been conducted in medicine for the needs of hand surgery and prosthetic design. Some of these studies refer to the categorization of six basic types of grasp: cylindrical, fingertip, hook, palmar, spherical and lateral [13]. These types of hand grasp, which can be seen in fig. 7. were associated with the shape of the objects to be grasped [14]. Another research [15] proposed that the choice of grasp type does not depend on the object shape and size, but on the task that is about to be performed. This theory reduced the grasp types into two distinctive categories, power grasp and precision grasp. The opposability of the thumb, provided by a mounting angle and rotation joint in its base allow the presented robotic hand to provide several basic grasps, shown in fig. 8.

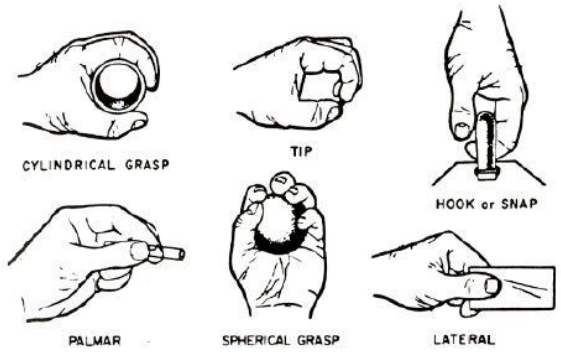


Fig. 7. Six basic types of grasp



a) b)



c) d)

Fig. 8. a) Cylindrical grasp b) Lateral grasp c) Three fingered pinch d) Tip pinch

V. Conclusion

This paper presents the design of a low-cost underactuated anthropomorphic robotic hand. The hand has the ability of passive adaptation to the shape of a grasped object due to the proposed tendon driven mechanism and torsional springs in finger joints. The proposed design is biologically inspired and due to the possibilities of 3D printing technology it is very compact with a complex anthropomorphic shape. Complete actuation system is embedded in the palm, as well as the power and control electronics. The hand has 16 joints and only 5 actuators, which makes it highly underactuated. Additional rotational joint in the thumb base and the mounting angle of the thumb provide the hand with the ability to execute several basic grasps, suitable for great number of objects from ours environment. Custom made force sensors integrated in finger links enable the grasping of different materials and fragile objects as well. Some ideas for future work include application of machine learning techniques for learning hand dynamics and for choosing appropriate type of grasp based on information from vision system.

Appendix

Acknowledgment

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